Computer Organization and Architecture (CSL251)
LAB-30.03.2026
2025-26 W

Paged Memory Organization-Revision

Registers

- RISC-V architecture provides 31 user modifiable general-purpose registersy, x1 to x31.
- It also provides an additional read-only register x0, which is hard-wired to zero.

Register Name	ABI Name	Description
x0	zero	Hard-Wired Zero
x1	ra	Return Address
x2	sp	Stack Pointer
x3	gp	Global Pointer
x4	tp	Thread Pointer
x5	t0	Temporary/Alternate Link Register
x6-7	t1-t2	Temporary Register
x8	s0/fp	Saved Register (Frame Pointer)
x9	s1	Saved Register
x10-11	a0-a1	Function Argument/Return Value Registers
x12-17	a2-a7	Function Argument Registers
x18-27	s2-s11	Saved Registers
x28-31	t3-t6	Temporary Registers

Privilege modes

Privilege	Value	Encoding	Abbreviation
User mode	0	00	U
Machine mode	3	11	M
Supervisor mode	1	01	S

- Each mode is represented by a value and encoding stored in CSR registers.

Paged Memory Organization

- Memory management technique used by operating systems to efficiently manage memory.

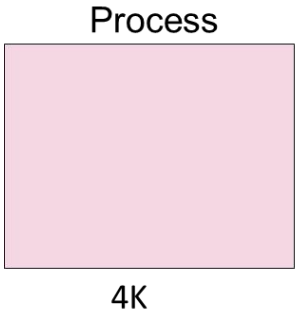
Key Idea

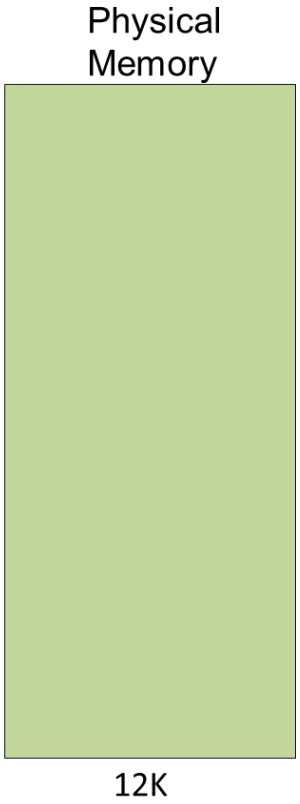
- The virtual memory of a program is divided into fixed-size blocks called pages.
- The physical memory is divided into blocks of the same size called frames.
- A page table is used to map pages to frames.

How It Works

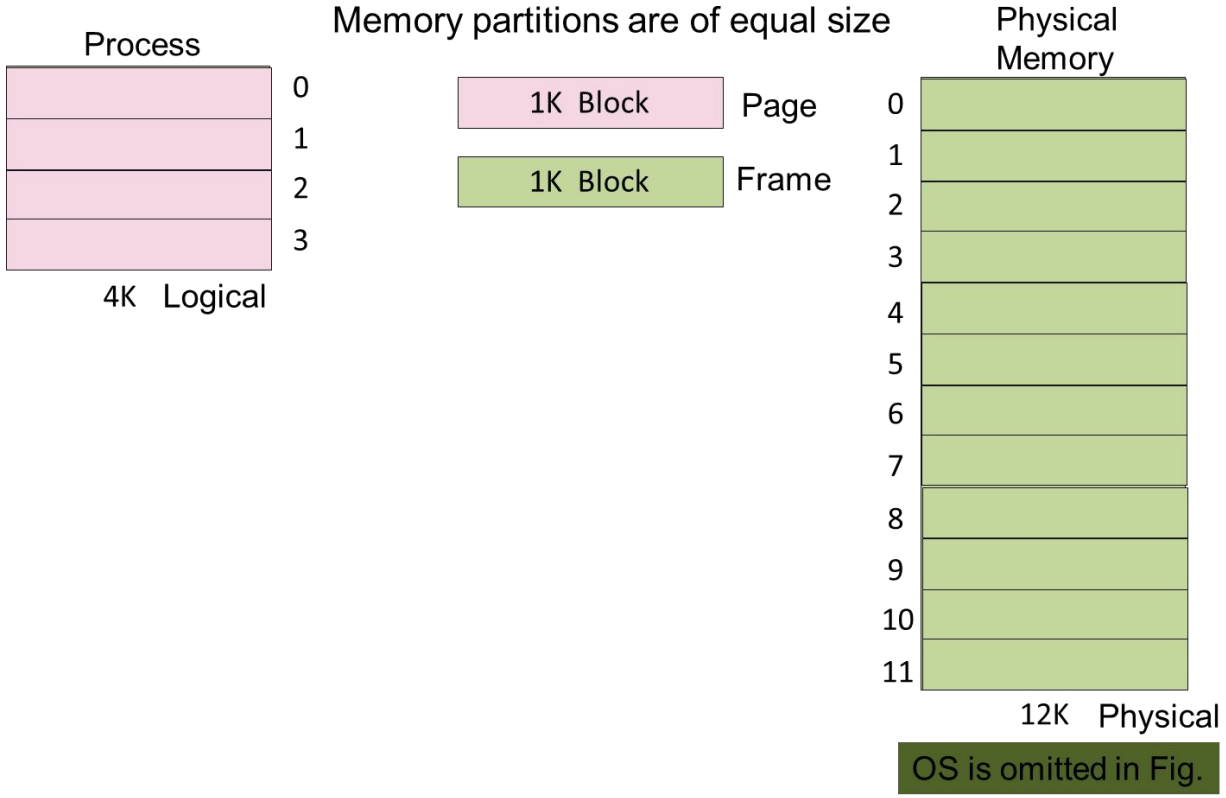
- CPU generates a virtual address.
- The page table converts it into a physical address.
- The data is then accessed from physical memory.

Paged Memory Organization

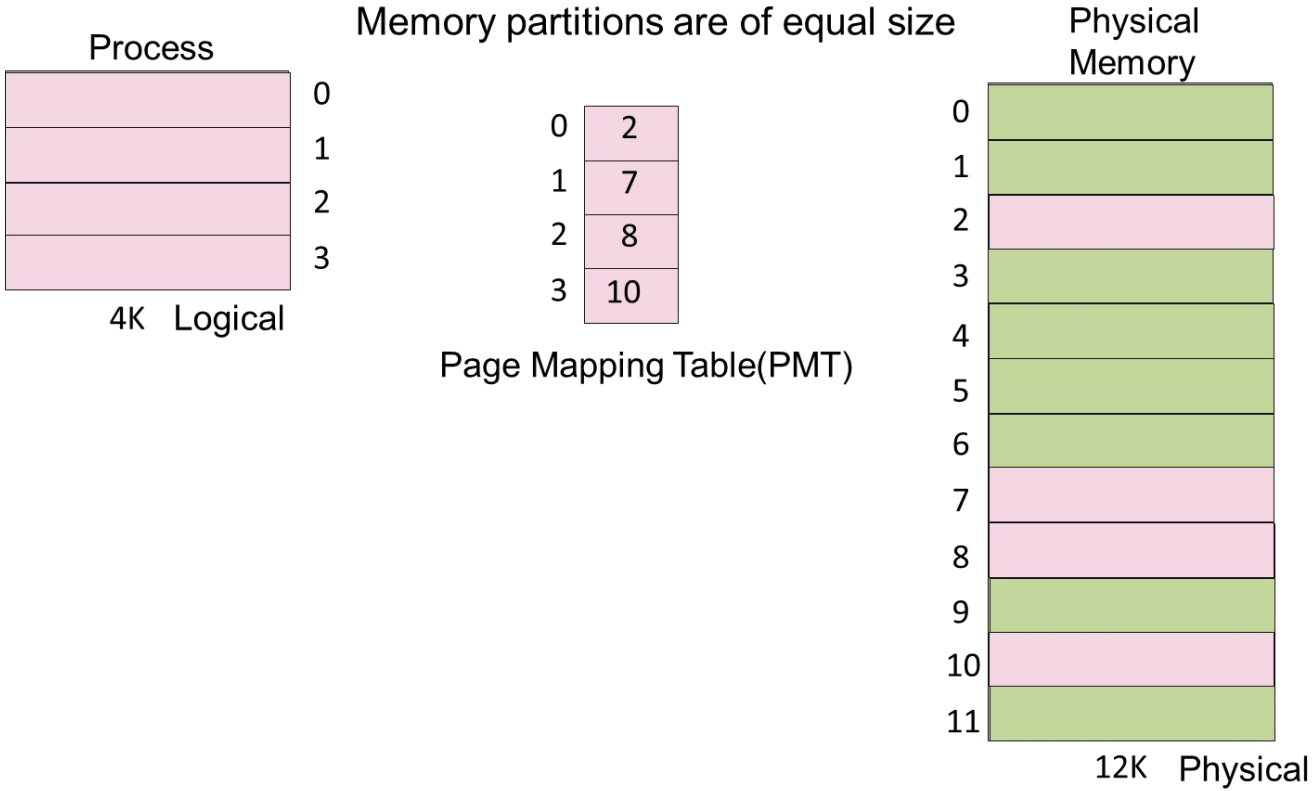




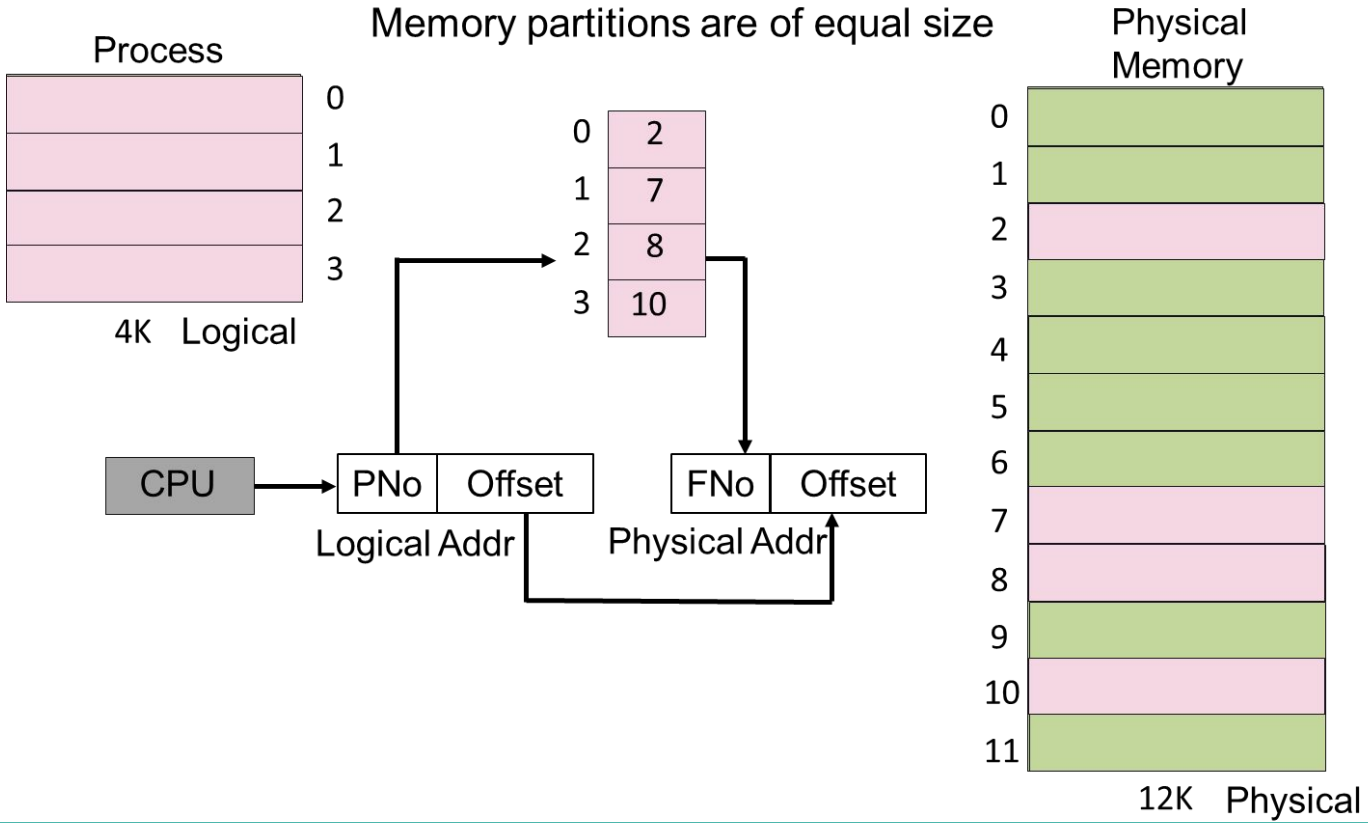
Paged Memory Organization



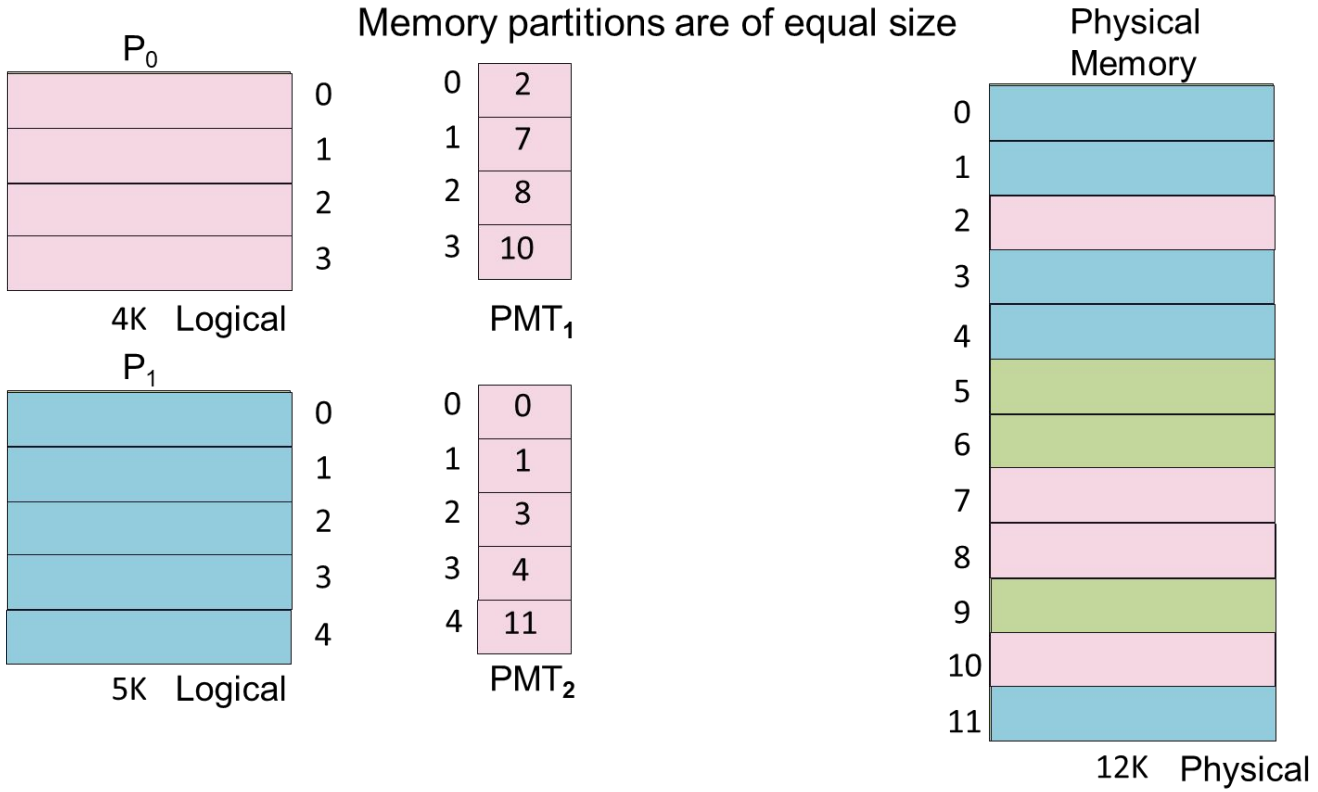
Paged Memory Organization



Paged Memory Organization



Paged Memory Organization



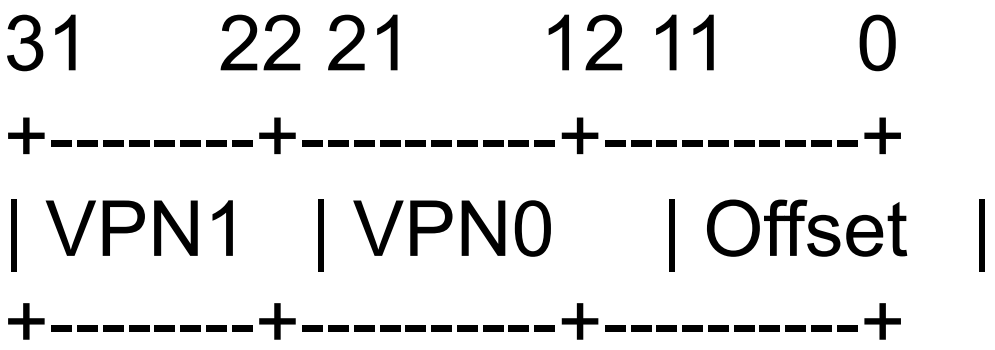
Paged Memory Organization

- Each entry in the page table contains:

Field	Meaning
PPN	Physical Page Number
V	Valid bit
R	Read permission
W	Write permission
X	Execute permission
A	Accessed bit
D	Dirty bit

Sv32 Paging in RISC-V

- 32-bit virtual address
- Two-level page table
- Page size = 4 KB



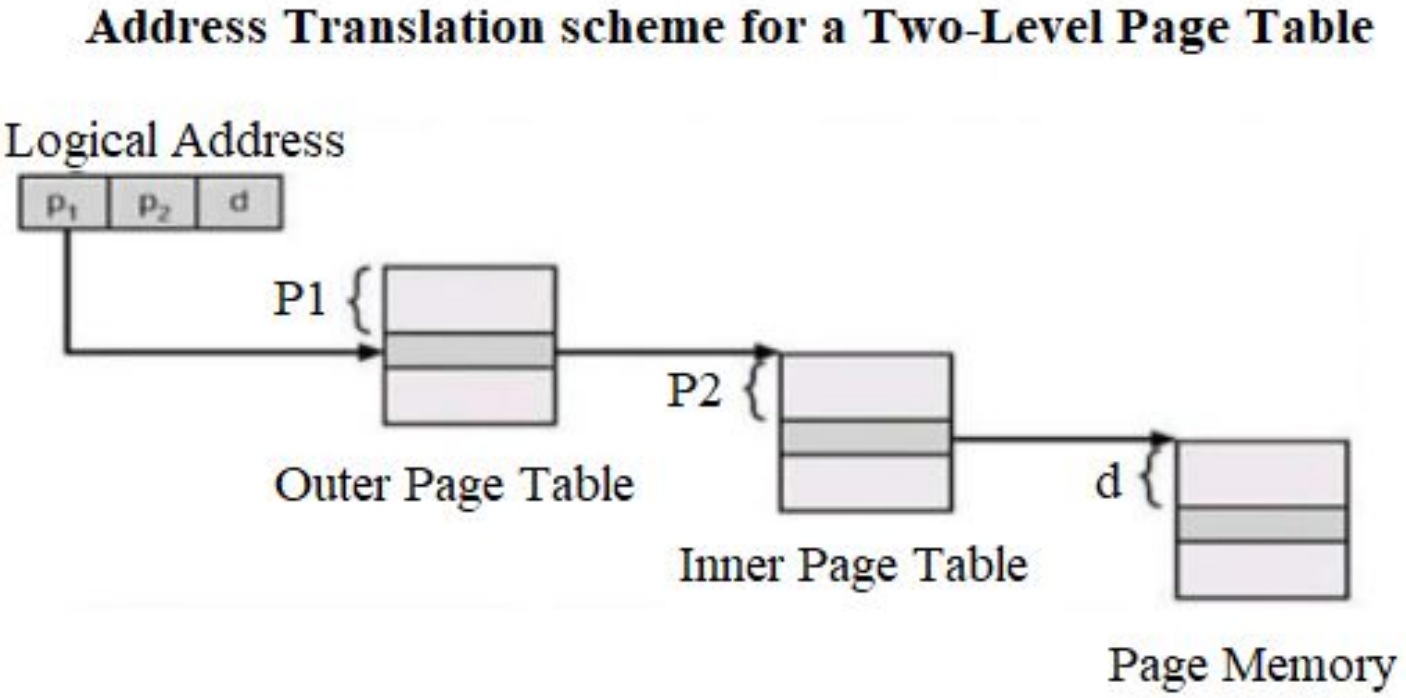
Where:

VPN1 → Level-1 page table

VPN0 → Level-0 page table

Offset → Byte inside page

Sv32 Paging in RISC-V



Two Level Paging in RISC-V

Machine Mode

↓

Supervisor Mode

↓

Create Page Table

↓

Enable Paging

↓

User Mode

↓

Execute Program

SATP(Supervisor Address Translation and Protection)

- satp is a control register in RISC-V that enables virtual memory paging.

It stores:

- **Paging mode-** Paging mode (Bare / Sv32)
- **Page table base address-** Physical Page Number of root page table

When the processor reads the satp register, it knows:

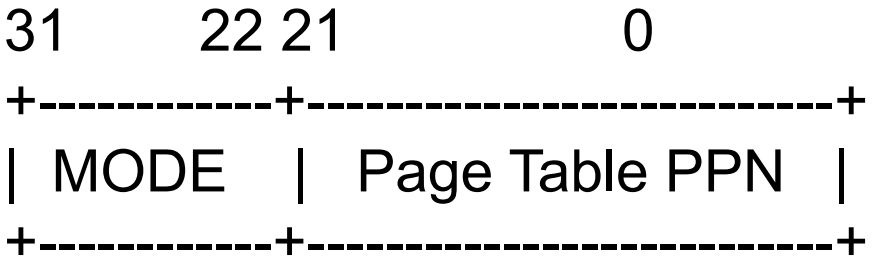
How to translate virtual addresses

and

Where the page table is located

SATP(Supervisor Address Translation and Protection)

SATP Structure (Sv32)



MODE	Meaning
0	Bare (no paging)
1	Sv32 paging

SATP(Supervisor Address Translation and Protection)

If program sets:

satp = 0x80081000

Binary breakdown:

1 00000000000010000001000000000000

↑

MODE

- The remaining bits represent the Physical Page Number (PPN) of the root page table.

PPN = 0x81000

SATP(Supervisor Address Translation and Protection)

- **Meaning:**

Paging Mode = Sv32

Root Page Table Address = 0x81000000

- **Because:**

PPN × Page Size

0x81000 × 0x1000 = 0x81000000

SATP(Supervisor Address Translation and Protection)

- **When this instruction runs:**

  csrw satp, t1

- **The processor loads:**

  satp = 0x80081000

- **Now the processor knows:**

  Paging mode → Sv32

  Root page table → 0x81000000

sfence.vma Instruction (TLB Flush)

- RISC-V instruction used to synchronize virtual memory changes.
- Clears cached address translations stored in the TLB (Translation Lookaside Buffer).
- The CPU stores recent Virtual Address → Physical Address mappings in the TLB for faster access.
- When the page table changes, the old mappings in the TLB may become incorrect.
- So we use:

sfence.vma

- to flush the old translations.

Enabling Paging

- Sfence.vma
  - Clears old address translations
- csrw satp
  - Loads page table configuration
    - satp = 0x80081000
  - Meaning:
    - MODE = Sv32
    - Root page table = 0x81000000
- Second sfence.vma
  - Ensures new translations are used

Page Table Entry Bits

- Sfence.vma
  - Clears old address translations
- csrw satp
  - Loads page table configuration
    - satp = 0x80081000
  - Meaning:
    - MODE = Sv32
    - Root page table = 0x81000000
- Second sfence.vma
  - Ensures new translations are used

Three-Level Paging

- Virtual addresses are large (39 bits in Sv39).
- Virtual Address is divided into parts:

VPN[2]	VPN[1]	VPN[0]	Page Offset
9 bits	9 bits	9 bits	12 bits

- Each level uses 9 bits to index a page table.
- Each page table contains 512 entries.

Three-Level Paging

Virtual Address

↓

Level 2 Page Table (Root)

↓

Level 1 Page Table

↓

Level 0 Page Table

↓

Physical Page

What is mstatus?

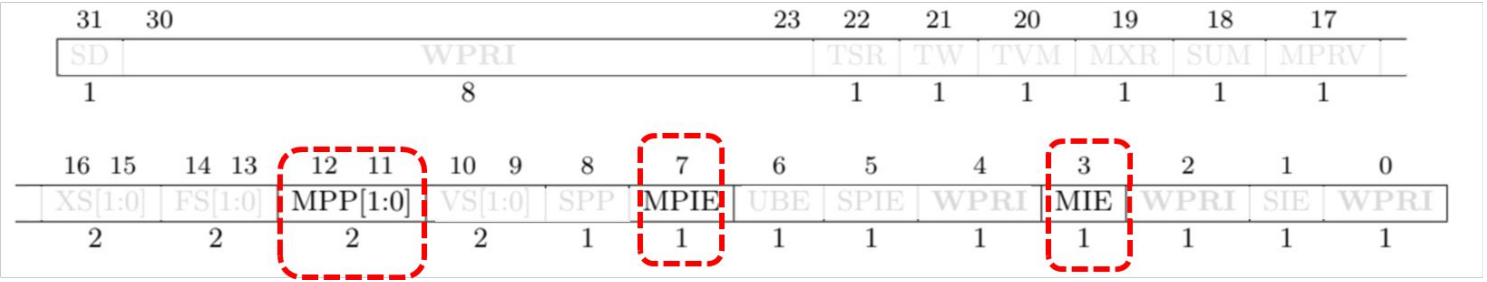
- mstatus = Machine Status Register

It controls:

- Privilege modes
- Interrupts
- Memory access behavior
- Floating-point/vector status

Interrupt Handling in RISC V (previous privilege mode)

4 • Stored in the CSR register: `mstatus`



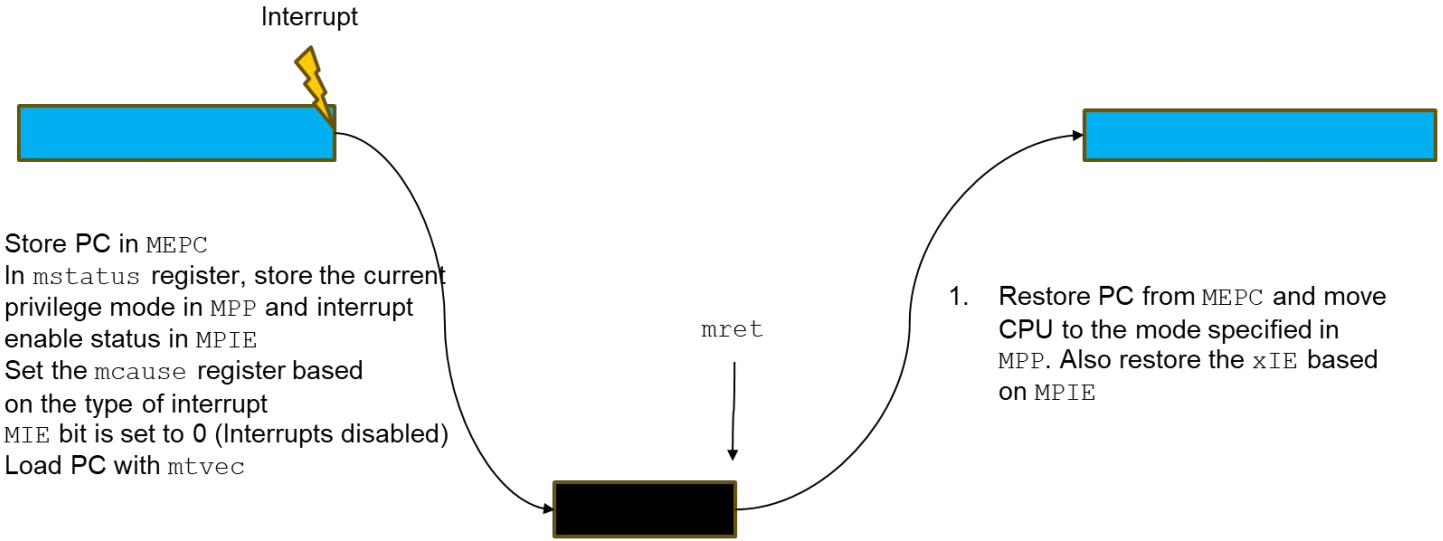
MPP: 0 → previous mode is user mode; 1 → previous mode is supervisor; 3 → previous mode is machine mode

MIE → If 1, enable interrupts in Machine mode;

MPIE → Previous mode interrupt enable status

In this course, we will restrict to the machine mode.
There are similar status bits, such as SPP, SPIE, etc. for supervisor mode.

Interrupt Handling in RISC V (Putting it all together)



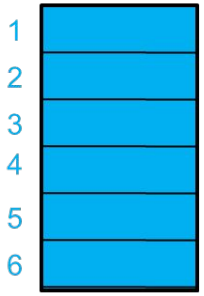
1. Store PC in `MEPC`
2. In `mstatus` register, store the current privilege mode in `MPP` and interrupt enable status in `MPIE`
3. Set the `mcause` register based on the type of interrupt
4. `MIE` bit is set to 0 (Interrupts disabled)
5. Load PC with `mtvec`

1. Restore PC from `MEPC` and move CPU to the mode specified in `MPP`. Also restore the `xIE` based on `MPIE`

Virtual Memory

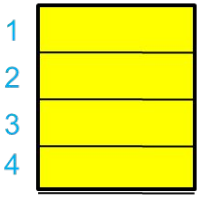


Blocks from Several processes can share pages in RAM simultaneously



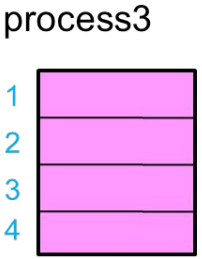
process page table

block	page frame
1	14
2	2
3	13
4	4
5	1
6	8



process page table

block	page frame
1	10
2	7
3	12
4	9

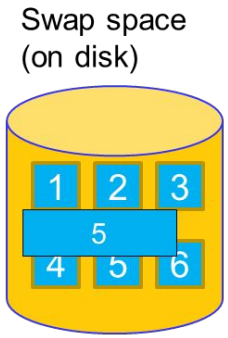


process page table

block	page frame
1	11
2	6
3	3
4	5

Demand Paging





process page table in RAM

block	page frame	P
1	14	1
2		0
3		0
4		0
5	1	0
6	8	1

present bit

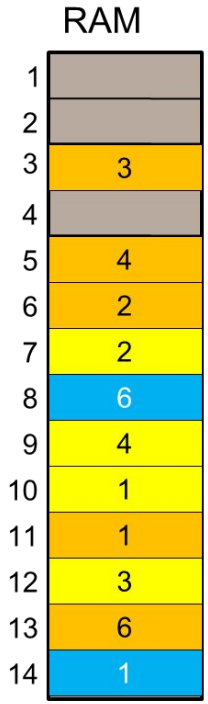
Pages are loaded from disk to RAM, only when needed.

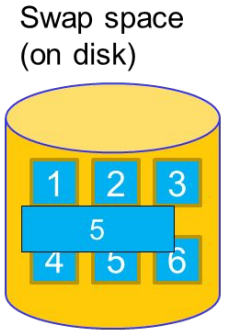
A ‘present bit’ in the page table indicates if the block is in RAM or not.

If (present bit = 1){ block in RAM}
else {block not in RAM}

If a page is accessed that is not present in RAM, the processor issues a page fault interrupt, triggering the OS to load the page into RAM and mark the present bit to 1

Demand Paging





process page table in RAM

block	page frame	P
1	14	1
2		0
3		0
4		0
5	1	0
6	8	1

present bit

Pages are loaded from disk to RAM, only when needed.

A 'present bit' in the page table indicates if the block is in RAM or not.

If (present bit = 1){ block in RAM}
else {block not in RAM}

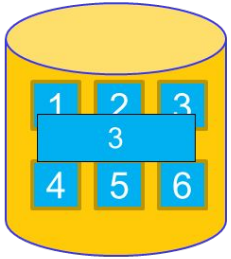
If a page is accessed that is not present in RAM, the processor issues a page fault interrupt, triggering the OS to load the page into RAM and mark the present bit to 1

Demand Paging

RAM

1	5
2	2
3	3
4	4
5	4
6	2
7	2
8	6
9	4
10	1
11	1
12	3
13	6
14	1

Swap space
(on disk)



process page table in RAM

block	page frame	P
1	14	0
2	2	1
3	14	1
4	4	1
5	1	1
6	8	1

If there are no pages free for a new block to be loaded, the OS makes a decision to remove another block from RAM.

This is based on a replacement policy, implemented in the OS.

Some replacement policies are
* First in first out
* Least recently used
* Least frequently used

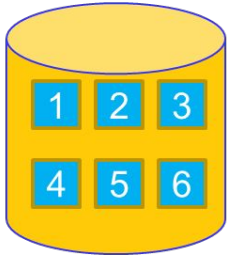
The replaced block **may** need to be written back to the swap (swap out)

Demand Paging

RAM

1	5
2	2
3	3
4	4
5	4
6	2
7	2
8	6
9	4
10	1
11	1
12	3
13	6
14	3

Swap space
(on disk)



process page table in RAM

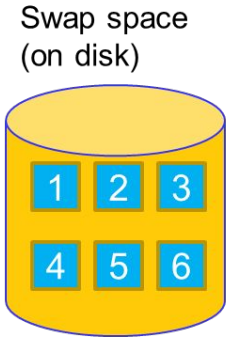
block	page frame	P	D
1	14	0	1
2	2	1	1
3	14	1	0
4	4	1	1
5	1	1	0
6	8	1	1

The **dirty bit**, in the page table indicates if a page needs to be written back to disk

If the dirty bit is 1, indicates the page needs to be written back to disk.

Demand Paging





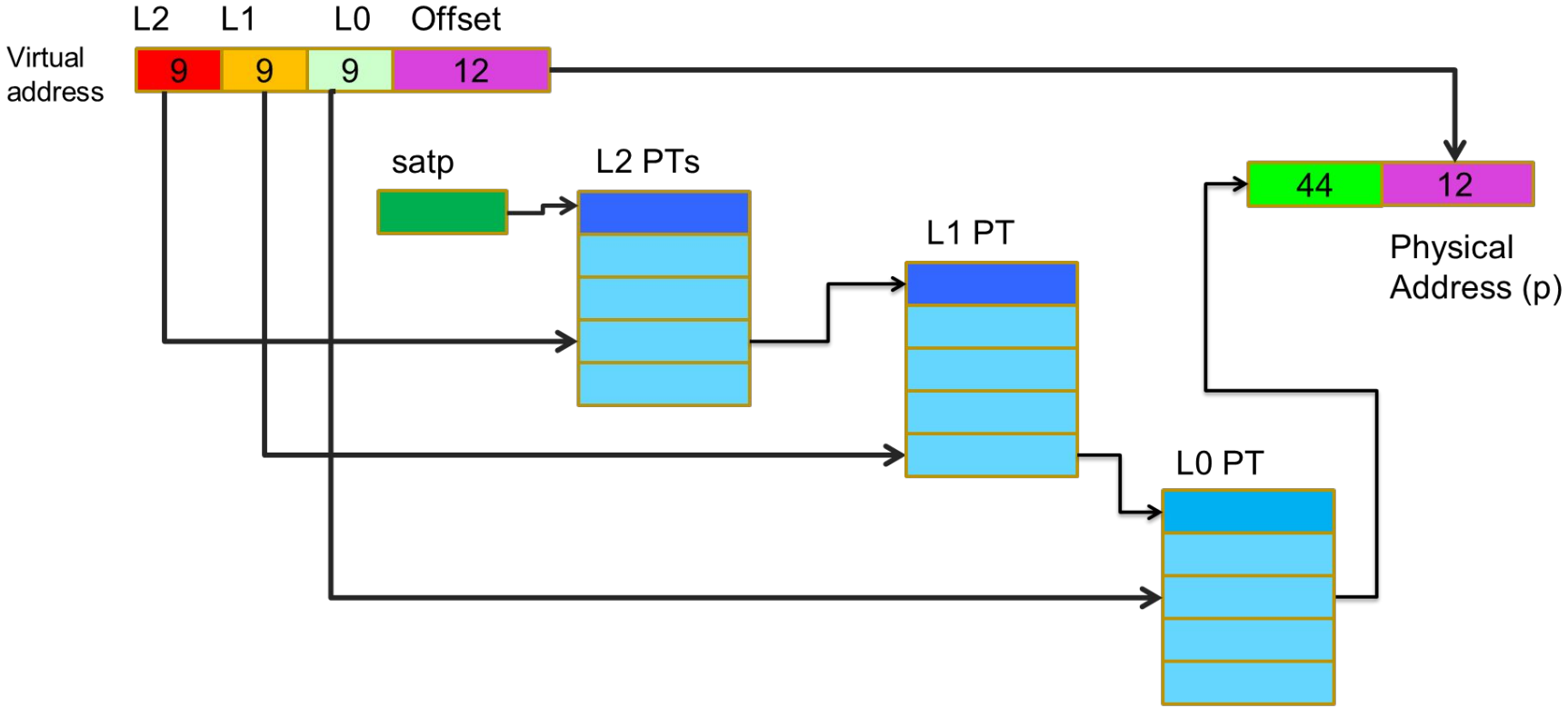
Protection bits, in the page table determine if
the page is executable, readonly, and accessible by a user process.

process page table in RAM

protection bits

block	page frame	P	D	
1	14	0	1	11
2	2	1	1	10
3	14	1	0	00
4	4	1	1	11
5	1	1	0	01
6	8	1	1	10

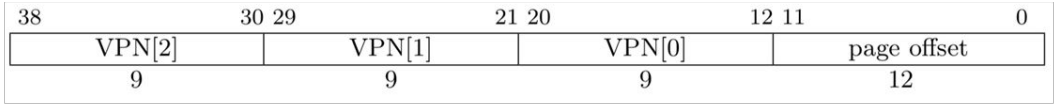
3 Level Page Translation



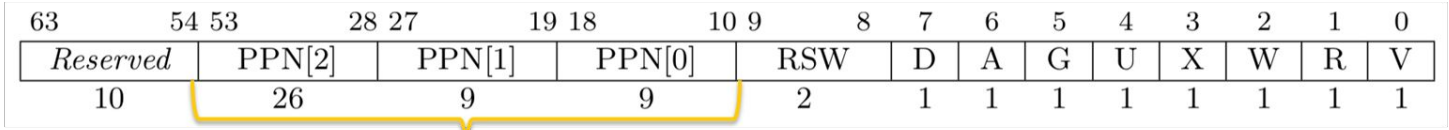
39-bit addressing in RISC V 64

- Three levels of page tables (can support 4KB, 2MB, 1GB pages)

Virtual address

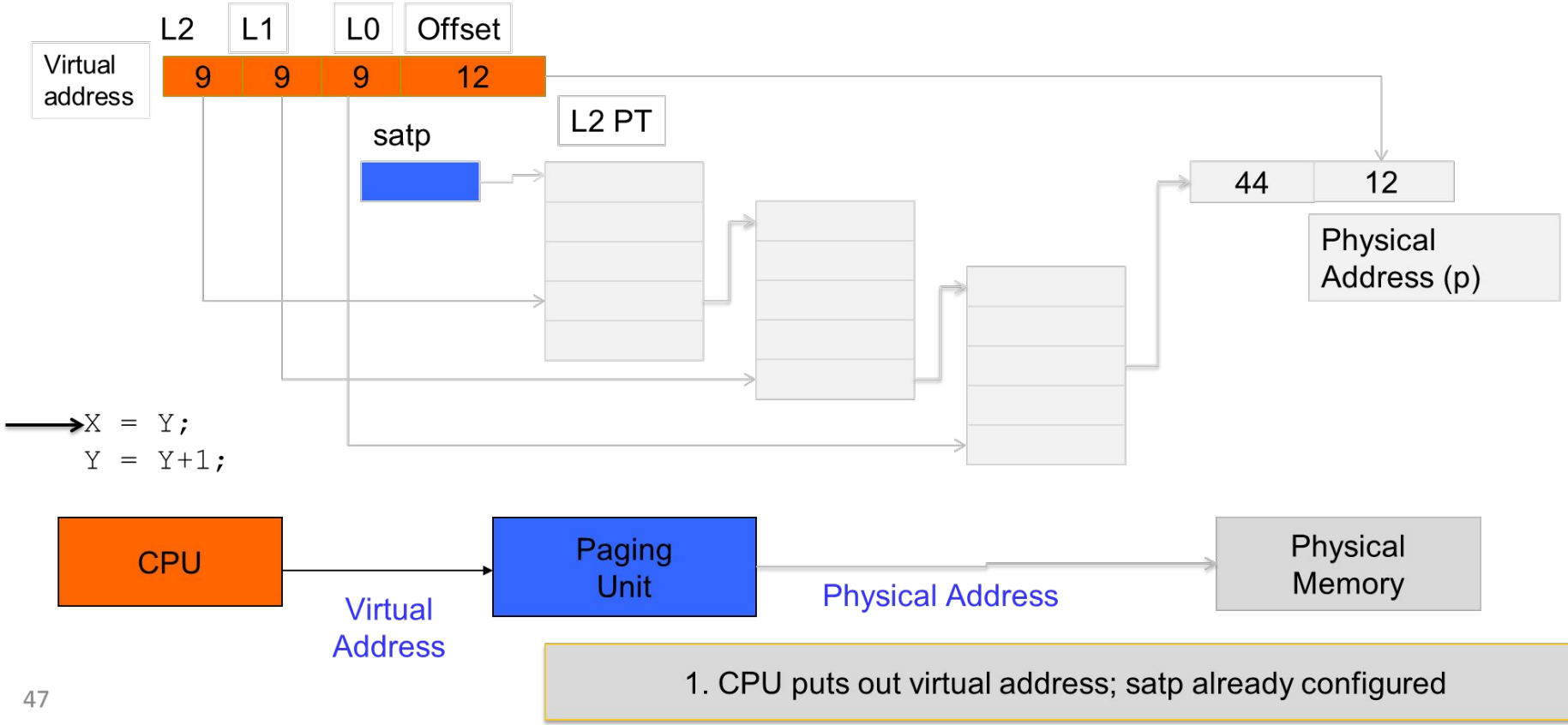


Page table entry

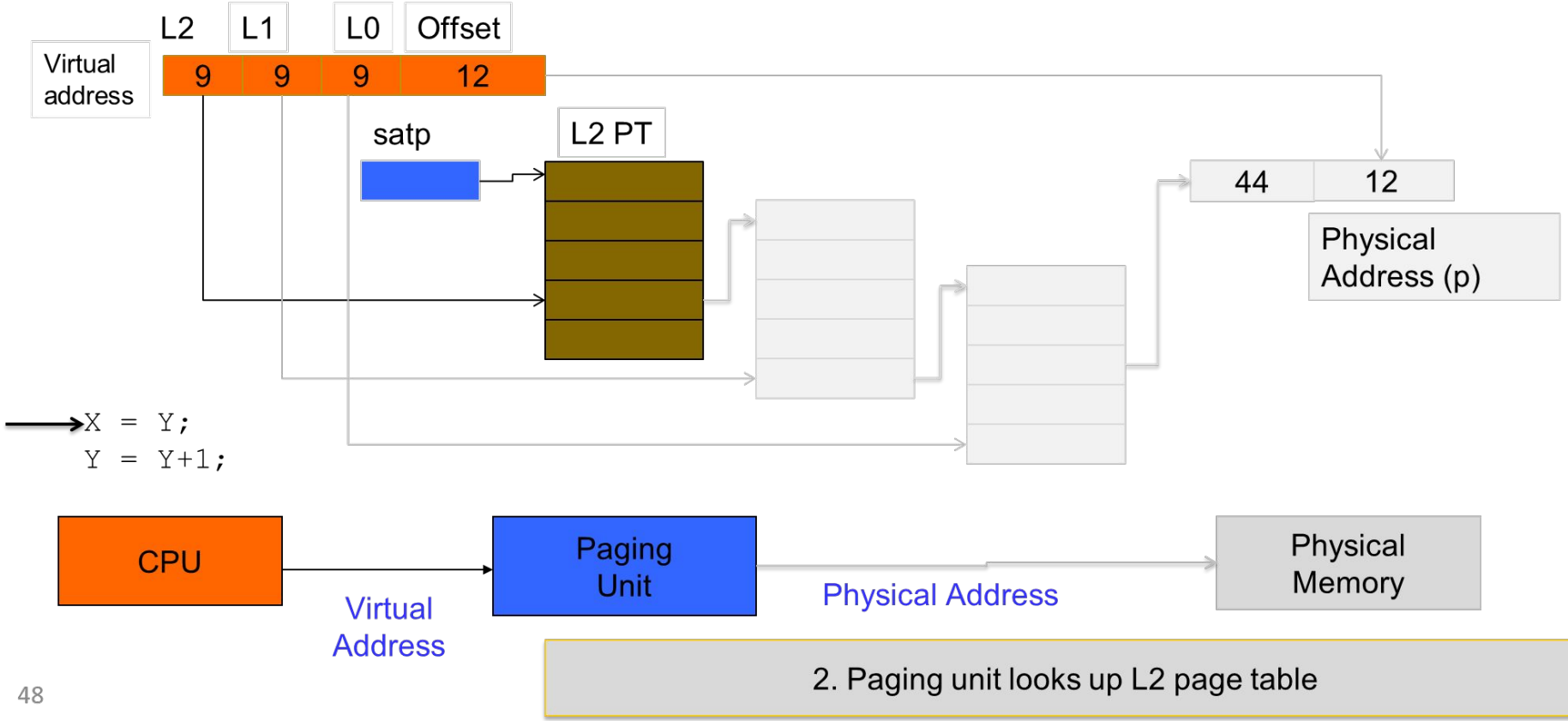


44-bit physical address

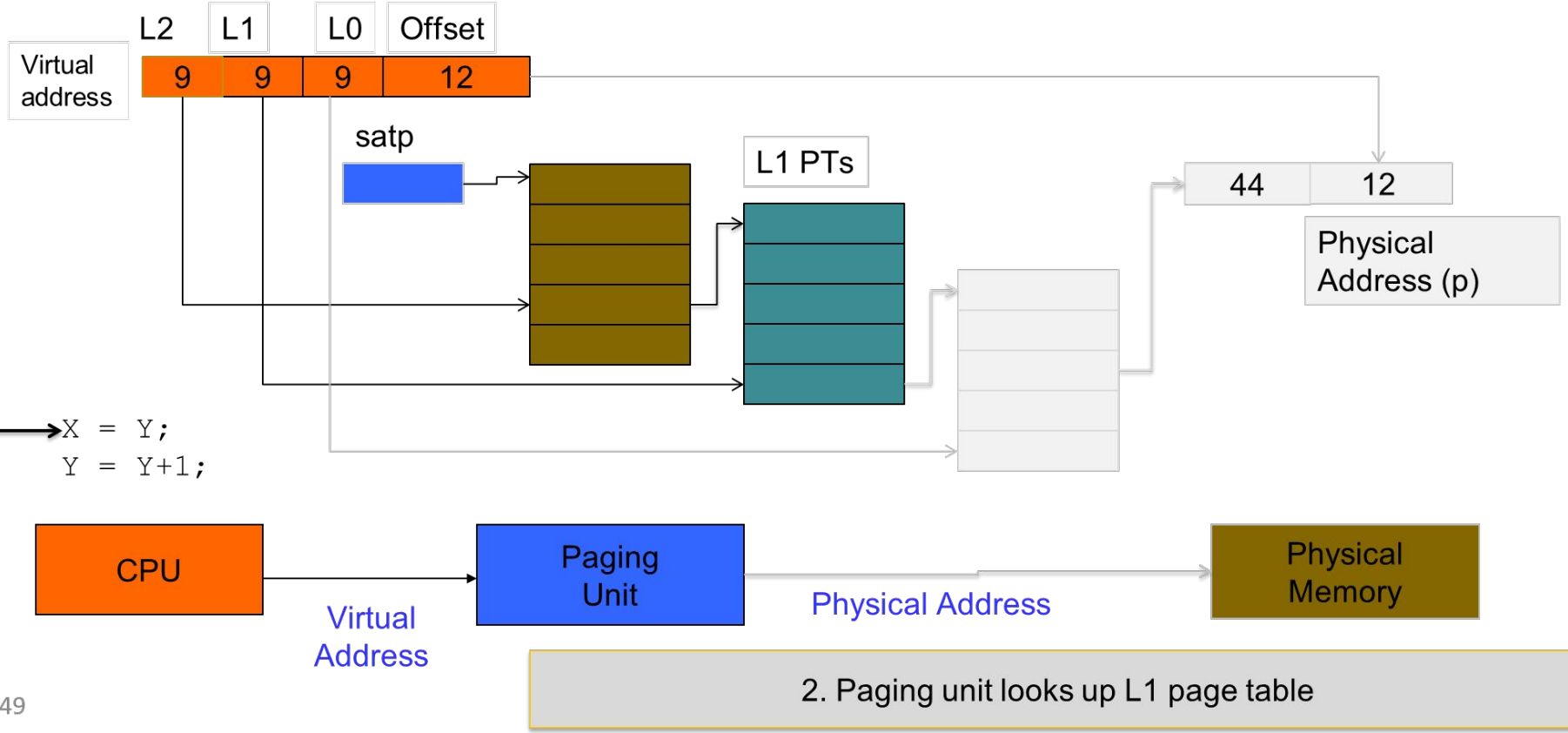
3 Level Page Translation



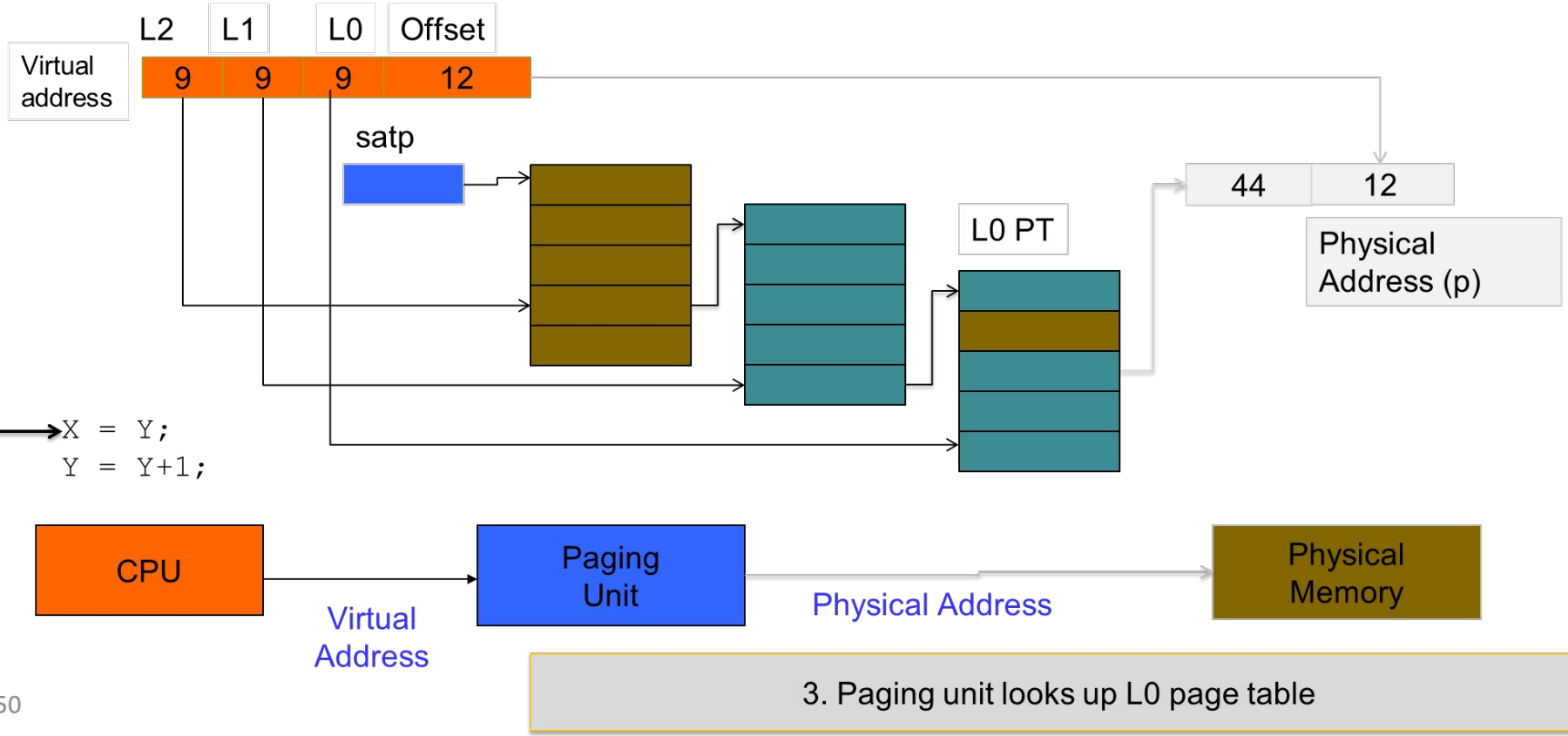
3 Level Page Translation



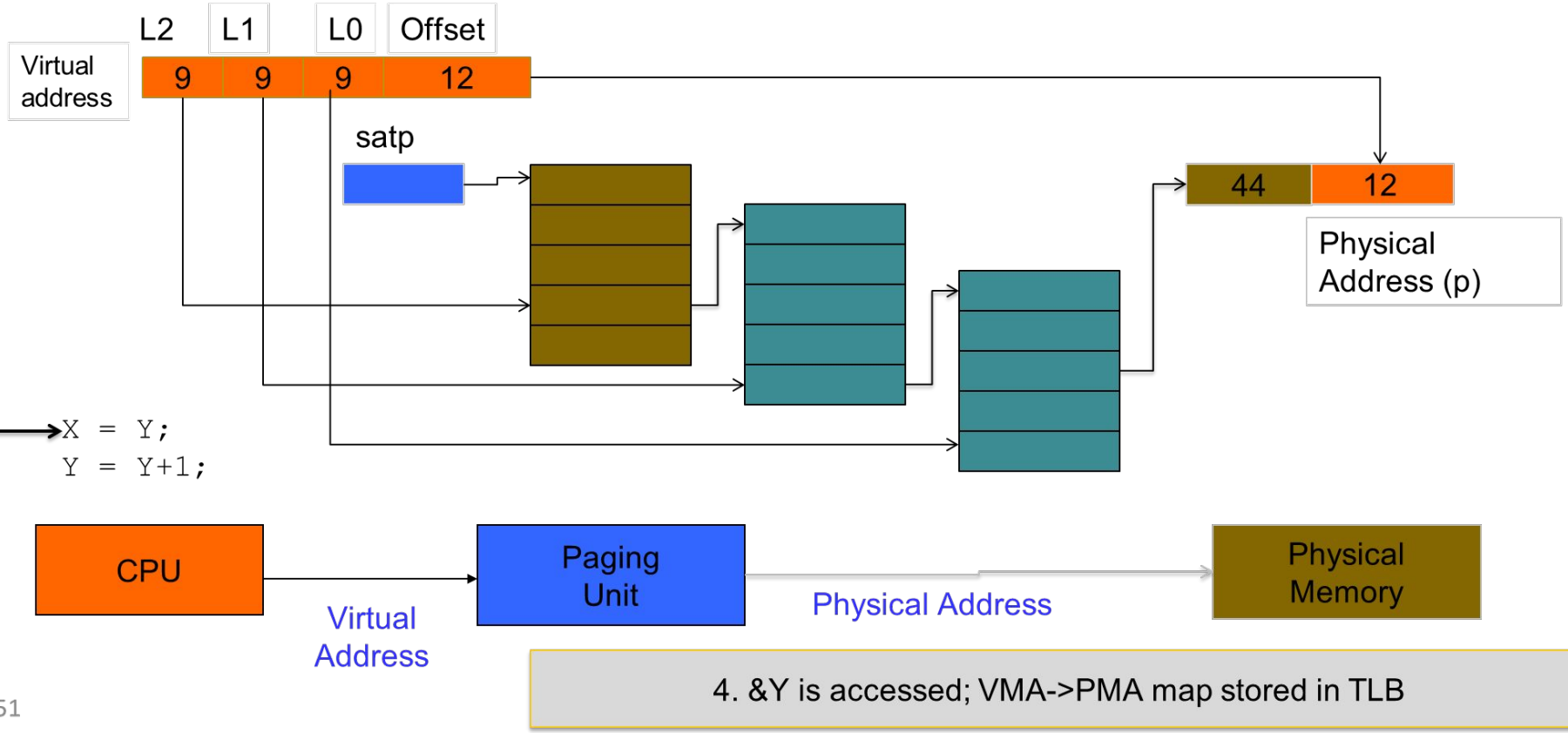
3 Level Page Translation



3 Level Page Translation



3 Level Page Translation



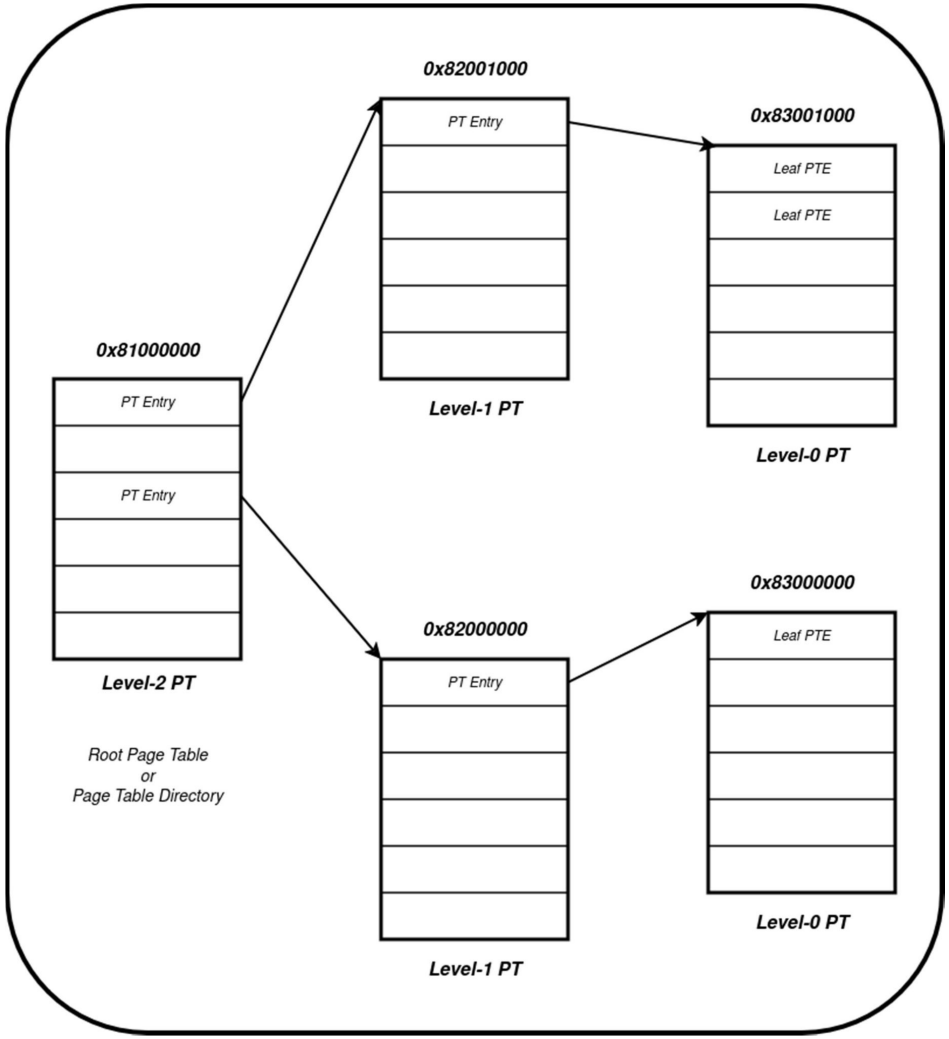
SATP(Supervisor Address Translation and Protection)

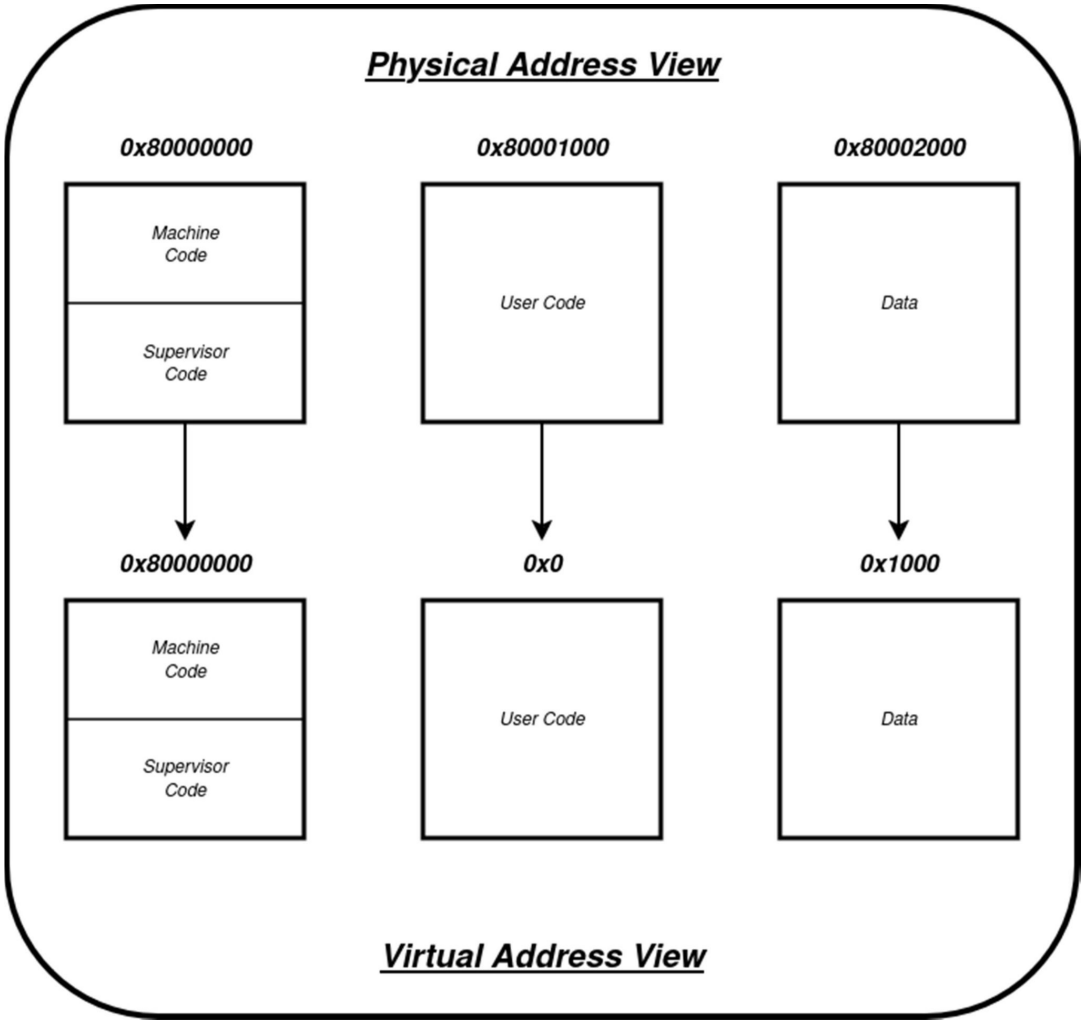
- satp is a control register in RISC-V that enables virtual memory paging.

It stores:

- **Paging mode-** Paging mode (Bare / Sv32)
- **Page table base address-** Physical Page Number of root page table
- **satp** = 0x8000000000081000

Bits	Field	Value
63–60	MODE	8 (Sv39)
59–44	ASID	0
43–0	PPN	0x81000


0x82001000
PT Entry
0x83001000
Leaf PTE
Leaf PTE
Level-1 PT
Level-0 PT
0x81000000
PT Entry
PT Entry
Level-2 PT
Root Page Table
or
Page Table Directory
0x83000000
0x82000000
Leaf PTE
PT Entry
Level-0 PT
Level-1 PT


Physical Address View
0x80000000
0x80001000
0x80002000
Machine Code
Supervisor Code
User Code
Data
0x80000000
0x0
0x1000
Machine Code
Supervisor Code
User Code
Data
Virtual Address View